Fermi's golden rule

Find the transition rate $R_{a\to b}$ for the perturbing Hamiltonian

$$H_1(\mathbf{r},t) = 2V(\mathbf{r})\cos(\omega t + \phi)$$

§1

Let $\Psi(\mathbf{r},t)$ be the following linear combination of two eigenstates.

$$\Psi(\mathbf{r},t) = c_a(t)\psi_a(\mathbf{r})\exp(-i\omega_a t) + c_b(t)\psi_b(\mathbf{r})\exp(-i\omega_b t)$$

We need to solve for $c_b(t)$ to find the transition rate $R_{a\to b}$.

Start with the Schrodinger equation

$$i\hbar\frac{\partial}{\partial t}\Psi(\mathbf{r},t) = H_0(\mathbf{r})\Psi(\mathbf{r},t) + H_1(\mathbf{r},t)\Psi(\mathbf{r},t)$$

where

$$H_0(\mathbf{r})\Psi(\mathbf{r},t) = E_a c_a(t)\psi_a(\mathbf{r})\exp(-i\omega_a t) + E_b c_b(t)\psi_b(\mathbf{r})\exp(-i\omega_b t)$$

Evaluate the time derivative.

$$\begin{aligned} i\hbar\frac{\partial}{\partial t}\Psi(\mathbf{r},t) = i\hbar\frac{\partial c_a(t)}{\partial t}\psi_a(\mathbf{r})\exp(-i\omega_a t) &+ \boxed{\hbar\omega_a c_a(t)\psi_a(\mathbf{r})\exp(-i\omega_a t)} \\ + i\hbar\frac{\partial c_b(t)}{\partial t}\psi_b(\mathbf{r})\exp(-i\omega_b t) &+ \boxed{\hbar\omega_b c_b(t)\psi_b(\mathbf{r})\exp(-i\omega_b t)} \\ &= H_0(\mathbf{r})\Psi(\mathbf{r},t) + H_1(\mathbf{r},t)\Psi(\mathbf{r},t) \end{aligned} \tag{1}$$

We have $\hbar\omega_a = E_a$ and $\hbar\omega_b = E_b$ hence the boxed terms cancel with $H_0(\mathbf{r})\Psi(\mathbf{r},t)$ leaving

$$i\hbar\frac{\partial c_a(t)}{\partial t}\psi_a(\mathbf{r})\exp(-i\omega_a t) + i\hbar\frac{\partial c_b(t)}{\partial t}\psi_b(\mathbf{r})\exp(-i\omega_b t) = H_1(\mathbf{r},t)\Psi(\mathbf{r},t)$$

Substitute for $H_1(\mathbf{r},t)$ and $\Psi(\mathbf{r},t)$.

$$\begin{aligned} i\hbar\frac{\partial c_a(t)}{\partial t}&\psi_a(\mathbf{r})\exp(-i\omega_a t) + i\hbar\frac{\partial c_b(t)}{\partial t}\psi_b(\mathbf{r})\exp(-i\omega_b t) \\ &= 2V(\mathbf{r})\cos(\omega t + \phi)\Big[c_a(t)\psi_a(\mathbf{r})\exp(-i\omega_a t) + c_b(t)\psi_b(\mathbf{r})\exp(-i\omega_b t)\Big] \end{aligned}$$

Apply an inner product with $\psi_b^*(\mathbf{r})$ to eliminate wave functions.

$$i\hbar\frac{\partial c_b(t)}{\partial t}\exp(-i\omega_b t) = 2\cos(\omega t + \phi)\Big[c_a(t)M_{21}\exp(-i\omega_a t) + c_b(t)M_{22}\exp(-i\omega_b t)\Big]$$

Symbols M_{21} and M_{22} are the matrix elements

$$M_{21} = \int \psi_b^*(\mathbf{r})V(\mathbf{r})\psi_a(\mathbf{r})\,d\mathbf{r}$$
$$M_{22} = \int \psi_b^*(\mathbf{r})V(\mathbf{r})\psi_b(\mathbf{r})\,d\mathbf{r}$$

Multiply both sides by $\exp(i\omega_b t)$.

$$i\hbar\frac{\partial c_b(t)}{\partial t} = 2\cos(\omega t + \phi)\Big[c_a(t)M_{21}\exp[i(\omega_b - \omega_a)t] + c_b(t)M_{22}\Big]$$

For time t near the origin use $c_a(t) \approx 1$ and $c_b(t) \approx 0$ to obtain

$$i\hbar\frac{\partial c_b(t)}{\partial t} = 2\cos(\omega t + \phi)M_{21}\exp[i(\omega_b - \omega_a)t]$$

Solve for $c_b(t)$ by integrating.

$$c_b(t) = \frac{2M_{21}}{i\hbar}\int_0^t \cos(\omega t' + \phi)\exp[i(\omega_b - \omega_a)t']\,dt'$$

In exponential form

$$c_b(t) = \frac{M_{21}}{i\hbar}\int_0^t \exp(i\omega t' + \phi)\exp[i(\omega_b - \omega_a)t']\,dt' + \frac{M_{21}}{i\hbar}\int_0^t \exp(-i\omega t' - \phi)\exp[i(\omega_b - \omega_a)t']\,dt'$$

The solution is

$$c_b(t) = -\frac{M_{21}}{\hbar}\,\frac{\exp[i(\omega_b - \omega_a + \omega)t] - 1}{\omega_b - \omega_a + \omega}\,\exp(i\phi) - \frac{M_{21}}{\hbar}\,\frac{\exp[i(\omega_b - \omega_a - \omega)t] - 1}{\omega_b - \omega_a - \omega}\,\exp(-i\phi) \quad (2)$$

For ω such that $\omega \approx \omega_b - \omega_a$ the second term dominates so discard the first term.

$$c_b(t) = -\frac{M_{21}}{\hbar}\,\frac{\exp[i(\omega_b - \omega_a - \omega)t] - 1}{\omega_b - \omega_a - \omega}\,\exp(-i\phi)$$

By the identity

$$\exp(ix) - 1 = 2i\sin\left(\frac{x}{2}\right)\exp\left(\frac{ix}{2}\right) \quad (3)$$

we have

$$c_b(t) = -\frac{2iM_{21}}{\hbar}\,\frac{\sin\left[\frac{1}{2}(\omega_b - \omega_a - \omega)t\right]\exp\left[\frac{1}{2}i(\omega_b - \omega_a - \omega)t\right]}{\omega_b - \omega_a - \omega}\,\exp(-i\phi)$$

Multiply numerator and denominator by $t/2$ to prepare for sinc function.

$$c_b(t) = -\frac{itM_{21}}{\hbar}\,\frac{\sin\left[\frac{1}{2}(\omega_b - \omega_a - \omega)t\right]\exp\left[\frac{1}{2}i(\omega_b - \omega_a - \omega)t\right]}{\frac{1}{2}(\omega_b - \omega_a - \omega)t}\,\exp(-i\phi)$$

Substitute $\operatorname{sinc}(x)$ for $\sin(x)/x$.

$$c_b(t) = -\frac{itM_{21}}{\hbar}\operatorname{sinc}\left[\frac{1}{2}(\omega_b - \omega_a - \omega)t\right]\exp\left[\frac{1}{2}i(\omega_b - \omega_a - \omega)t\right]\exp(-i\phi) \quad (4)$$

§2

Transition density P is the squared magnitude of $c_b(t)$.

$$P = |c_b(t)|^2 = \frac{t^2}{\hbar^2} |M_{21}|^2 \operatorname{sinc}^2 \left[\frac{1}{2}(\omega_b - \omega_a - \omega)t \right] \tag{5}$$

Total density P_{tot} is the integral of P over a small energy band.

$$P_{tot} = \frac{t^2}{\hbar^2} |M_{21}|^2 \int_{E-\epsilon}^{E+\epsilon} \operatorname{sinc}^2 \left[\frac{1}{2}(E'/\hbar - \omega)t \right] \rho(E')\, dE'$$

The center of the energy band is $E = \hbar(\omega_b - \omega_a)$. Symbol ρ is density of states.

Use the approximation $\rho(E') \approx \rho(\hbar\omega)$ to obtain

$$P_{tot} = \frac{t^2}{\hbar^2} |M_{21}|^2 \rho(\hbar\omega) \int_{E-\epsilon}^{E+\epsilon} \operatorname{sinc}^2 \left[\frac{1}{2}(E'/\hbar - \omega)t \right] dE'$$

Let y be a change of variable such that

$$y = \frac{1}{2}(E'/\hbar - \omega)t$$

From the definition of y we have

$$E' = \frac{2\hbar y}{t} + \hbar\omega$$

hence

$$dE' = \frac{2\hbar}{t}\, dy$$

The integral limits transform as

$$E \pm \epsilon \to \frac{1}{2}[(E \pm \epsilon)/\hbar - \omega]t = \frac{Et}{2\hbar} - \frac{\omega t}{2} \pm \frac{\epsilon t}{2\hbar} = \pm \frac{\epsilon t}{2\hbar}$$

After change of variable from E' to y we have (note $t^2/\hbar^2$ is reduced to $t/\hbar$)

$$P_{tot} = \frac{2t}{\hbar} |M_{21}|^2 \rho(\hbar\omega) \int_{-\epsilon t/2\hbar}^{\epsilon t/2\hbar} \operatorname{sinc}^2 y\, dy$$

Use the approximation

$$\int_{-\epsilon t/2\hbar}^{\epsilon t/2\hbar} \operatorname{sinc}^2 y\, dy \approx \int_{-\infty}^{\infty} \operatorname{sinc}^2 y\, dy = \pi$$

to obtain

$$P_{tot} = \frac{2\pi t}{\hbar} |M_{21}|^2 \rho(\hbar\omega)$$

The transition rate is the derivative of P_{tot}.

$$R_{a\to b} = \frac{d}{dt} P_{tot} = \frac{2\pi}{\hbar} |M_{21}|^2 \rho(\hbar\omega) \tag{6}$$

§3

We started with

$$H_1(\mathbf{r},t) = 2V(\mathbf{r})\cos(\omega t + \phi)$$

and obtained Fermi's golden rule (6). Consequently for $H_1(\mathbf{r},t)$ such that

$$H_1(\mathbf{r},t) = V(\mathbf{r})\cos(\omega t + \phi)$$

we would have

$$R_{a\to b} = \frac{2\pi}{\hbar}\left|\frac{M_{21}}{2}\right|^2 \rho(\hbar\omega)$$

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